

Your grade: 89%

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3.

Which of the following statements are true about the architecture and working of Generative Adversarial Networks (GANs)?

3 / 3 point
- ☐ GANs are primarily used for data classification tasks.

☐ The discriminator in GANs is responsible for generating new data.

☐ The generator in a GAN uses supervised learning to create data.

☒ GANs consist of two neural networks: a generator and a discriminator.

☒ Correct

Correct: GANs have two main components: the generator, which creates data, and the discriminator, which evaluates it.

☒ GANs use a technique called adversarial training to improve performance.

☐ Correct

Correct: Adversarial training is a key technique in GANs where the generator and discriminator are trained in opposition to each other.
3.

What is the primary purpose of the transpose convolution layer in the context of image processing in GANs?

3 / 3 point

☐ To convert an image to a generator.

☐ To reduce the dimensionality of an image.

☐ To apply random noise to an image.

☒ To increase the resolution of an image.

☐ Correct

Correct: The transpose convolution layer is used to increase the resolution of an image, which is essential in the generation phase of GANs.
3.

Select all statements that accurately describe the interdependent relationship between the generator and discriminator in a GAN network.

3 / 3 point

☐ Both networks learn from the same loss function.

☒ The generator and discriminator are trained simultaneously in a competitive setting.

☐ Correct

Correct: Both networks are trained together, with each trying to outmaneuver the other in a competitive manner.

☒ The discriminator's feedback helps the generator to improve its data generation.

☐ Correct

Correct: The generator uses feedback from the discriminator to enhance its ability to produce realistic data samples.

☐ The discriminator tries to fool the generator by providing deceptive feedback.

☐ The generator's main goal is to help the discriminator improve its accuracy.
4.

Which components are typically part of a fully connected GAN generator's architecture?

3 / 3 point

☒ Dropout layer

☐ Batch

☐ Conv

☒ Input layer

☐ Correct

Correct: The input layer is where the random noise is fed into the generation.

☐ Dropout layers

☐ Pooling layers

☒ Hidden layers

☐ Correct

Correct: Hidden layers process the input through various transformations.

☐ Recurrent layers
5.

Which of the following activation functions is most commonly used in the discriminator model of a GAN?

3 point

☒ Leaky ReLU

☐ Sigmoid

☐ Tanh (Hyperbolic Tangent)

☐ ReLU (Rectified Linear Unit)

☒ Incorrect

Leaky ReLU is sometimes used in the discriminator model, but it's not the most common activation function.
4.

What is the primary purpose of compiling the discriminator model in a GAN?

3 / 3 point

☐ To update the weights of the compiled GAN model.

☐ To reduce overfitting in the GAN model.

☐ To generate new data samples.

☒ To differentiate between real and generated data.

☐ Correct

Correct: The discriminator's primary role is to distinguish between real and generated data, which helps train the generator.
1.

Which of the following is the primary purpose of generating random noise in the training of a GAN?

3 / 3 point

☐ To preprocess the real images before feeding them to the discriminator.

☐ To initialize the weights of the discriminator model.

☐ To generate binary one-hot encoded labels for real and fake images.

☒ To create input for the generator to produce fake images.

☐ Correct

Correct: Generating random noise is essential for the generator to create fake images.
6.

Which of the following steps are involved in debugging the training process in a GAN network?

3 / 3 point

☐ Reviewing the loss on each method without checking intermediate outputs.

☒ Using the predict function to visualize generated images.

☐ Correct

Correct: Using the predict function allows you to visualize the images generated by the GAN.

☒ Adjusting the learning rate during training.

☐ Correct

Correct: Adjusting the learning rate can help improve training stability and performance.

☐ Ignoring any discrepancies between real and generated images.

☒ Printing out the loss values during training.

☐ Correct

Correct: Printing loss values helps track the performance of the model during training.
9.

Which of the following libraries is commonly used to plot graphs in Python for visualizing accuracy and loss?

3 / 3 point

☒ matplotlib

☐ numpy

☐ sklearn

☐ pandas

☐ Correct

Correct: Matplotlib is a widely used library for plotting graphs and visualizing data in Python.
10.

Which of the following steps is NOT required to plot sample images in a 4x4 matrix using a GAN generator model?

3 / 3 point

☒ Training the GAN generator model.

☐ Passing random vectors into the model for prediction.

☐ Generating random noise vectors.

☐ Displaying the generated images.

☐ Correct

Correct: While training the GAN generator model is crucial, it is not a step directly involved in plotting sample images.
11.

Which steps are involved in saving a generative network model?

8.8 / 3 point

☐ Loading the dataset into memory.

☐ Saving the generator state.

☐ Compiling the model.

☒ Serializing the model structure.

☐ Correct

Correct: Serializing the model structure is a critical step in saving the model.

☒ Saving the model weights.

☐ Correct

Correct: Saving the model weights is essential for preserving the trained parameters.

☐ No action is required.

Yes, but not all of the correct answers.
12.

What is the primary purpose of reshaping a noise vector into a 2D tensor before feeding it to a DCGAN generator?

3 / 3 point

☐ To enhance the quality of the generated images.

☐ To ensure the noise vector contains a wide range of frequencies.

☐ To reduce the computational complexity of the generation process.

☒ To match the input dimensions expected by the convolutional layers.

☐ Correct

Correct: Reshaping the noise vector ensures that it matches the input dimensions expected by the convolutional layers in the DCGAN generator, facilitating proper training and generation.
13.

What is the first step in converting a random vector into a 256x256 output image in a Generative Adversarial Network (GAN)?

3 / 3 point

☐ Adjusting the learning rate of the generator.

☐ Normalizing the random vector to the range [-1, 1].

☒ Inputting the random vector into the generator network.

☐ Using the discriminator network to refine the random vector.

☐ Correct

Correct: The process begins by inputting the random vector into the generator network. This vector will then be transformed through several layers to produce the final image.
14.

In the context of a discriminator model for GANs, what is the main purpose of implementing convolution layers?

3 / 3 point

☐ To increase the number of channels in the input image.

☐ To reduce the size of the input image to a single scalar value.

☐ To reduce the need for activation functions in the network.

☒ To extract features from the input image at various scales and locations.

☐ Correct

Correct: Convolution layers are essential for extracting features from the input image at different scales and locations, which helps the discriminator in distinguishing between real and generated images.
15.

What is the main purpose of combining labels with the generator function in a conditional GAN?

3 / 3 point

☐ To increase the overall training time of the GAN.

☒ To allow the GAN to generate data conditioned on specific labels.

☐ To make the GAN architecture more complex.

☐ To reduce the discriminator's error rate.

☐ Correct

Correct: Combining labels with the generator function enables the GAN to generate data that is conditioned on specific labels, which allows for more controlled and targeted data generation.
16.

Which of the following is a common issue that can occur during the training process of GANs?

3 / 3 point

☐ Underparameterization

☒ Mode collapse

☐ Overfitting

☐ Vanishing gradients

☐ Correct

Correct: Mode collapse occurs when the generator produces limited varieties of outputs, leading to poor model performance.
17.

What is the main difference between sequential and functional Keras models?

3 / 3 point

☐ Sequential models support only convolutional layers, while functional models support both convolutional and recurrent layers.

☒ Functional models are used exclusively for Generative Adversarial Networks (GANs), while sequential models are used for all other types of neural networks.

☐ Sequential models allow you to create models layer by layer in a linear stack, whereas functional models allow more flexibility, including branching and merging layers.

☐ Sequential models can only be used with TensorFlow, while functional models can be used with both TensorFlow and PyTorch.

☐ Correct

Correct: The sequential API is limited to linear stacks, while the functional API supports more complex architectures.
18.

Which of the following steps is involved in creating a random noise vector for a GAN model?

3 / 3 point

☒ Generating random vectors from a uniform distribution.

☐ Combining random images to form a response.

☐ Applying a Fourier transform to random data.

☐ Using the output of a ReLU activation function.

☐ Correct

Correct: This is a common method for creating random noise vectors for GAN models.
19.

Which of the following best describes the process of training the generator model in a Generative Adversarial Network (GAN)?

3 point

☐ The generator directly competes with other generators to improve.

☐ The generator creates fake data, which is then validated by the discriminator.

☐ The generator updates its weights based on real data labels.

☒ The generator model is trained by adjusting its parameters to minimize the loss of the combined model.

☒ Incorrect

Close, but not exactly. Focus on how the generator learns from the discriminator's feedback.
20.

Which of the following methods is used to create ordered sample labels in a dataset?

3 / 3 point

☐ Assigning labels based on the sample's size.

☐ Using the sample size to assign labels.

☒ Assigning consecutive integers to each class.

☐ Using random numbers for each sample.

☐ Correct

Correct: Assigning consecutive integers to each class is a common method to create ordered sample labels in a dataset.